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Final Report:
NYC Airbnb Predictive Modeling

Problem Statement

Airbnb hosts face challenges in understanding the market dynamics and optimizing their listings to maximize occupancy rates and rental income. The area of focus can be broken down into market entry strategy, pricing optimization, and demand forecasting.

Background

Airbnbs have become a popular option for both short- and long-term stays in New York City, driven by high hotel prices and the appeal of a more local experience. As a result, thousands of listings exist across the five boroughs, attracting many hosts looking to enter this lucrative market. However, new and current hosts often struggle to determine fair and competitive prices for their properties. This proposal aims to develop a predictive model to help hosts make informed, data-driven pricing decisions in the NYC Airbnb market.

Approach

We will begin by collecting historical Airbnb listing data for NYC, including pricing, property attributes, and neighborhood characteristics. After cleaning and preparing the data, we will perform exploratory analysis to uncover key correlations, emerging trends, and factors that influence listing performance—such as neighborhood demand and pricing patterns. Guided by our three focus areas, we will develop a regression model to support market entry by predicting rental prices based on location, amenities, and market conditions. For pricing optimization, we will benchmark competitor

rates and incorporate real-time supply-and-demand signals. Finally, we will build demand-forecasting models that project future pricing and occupancy trends using historical data and external influencing factors.

Data Wrangling

The dataset used in this analysis was the New York City Airbnb Open Data obtained from Kaggle, comprising over 48,000 listings from 2019. Prior to modeling, extensive data wrangling was performed to ensure data quality, consistency, and suitability for analysis.

Initial data analysis was conducted to examine variable types, summary statistics, and patterns of missingness. Features deemed non-informative for predictive modeling—such as host identifiers and host names—were removed, as they provided no meaningful explanatory value and could introduce unnecessary noise into the models.

Missing data were then addressed. Two variables, *last_review* and *reviews_per_month*, contained a substantial number of missing values. These missing entries were interpreted as listings without prior reviews and were retained by imputing appropriate default values, preserving the size and representativeness of the dataset.

The price variable was then closely examined for anomalies. Listings with invalid or extreme prices were removed to reduce distortion and improve model stability. In addition, the price variable was log-transformed to address its strong right skew and to better approximate a normal distribution. This transformation helps stabilize variance, reduces the influence of extreme values, and improves the performance and interpretability of regression-based models.

Together, these data wrangling steps produced a clean, well-structured dataset and established a strong foundation for exploratory analysis, feature engineering, and predictive modeling.

Exploratory Data Analysis

Exploratory data analysis was conducted to characterize the underlying structure of the dataset and identify key relationships between listing attributes and pricing behavior. Several univariate and multivariate visualizations were used to assess distributional patterns, spatial variation, and feature relevance prior to model development.

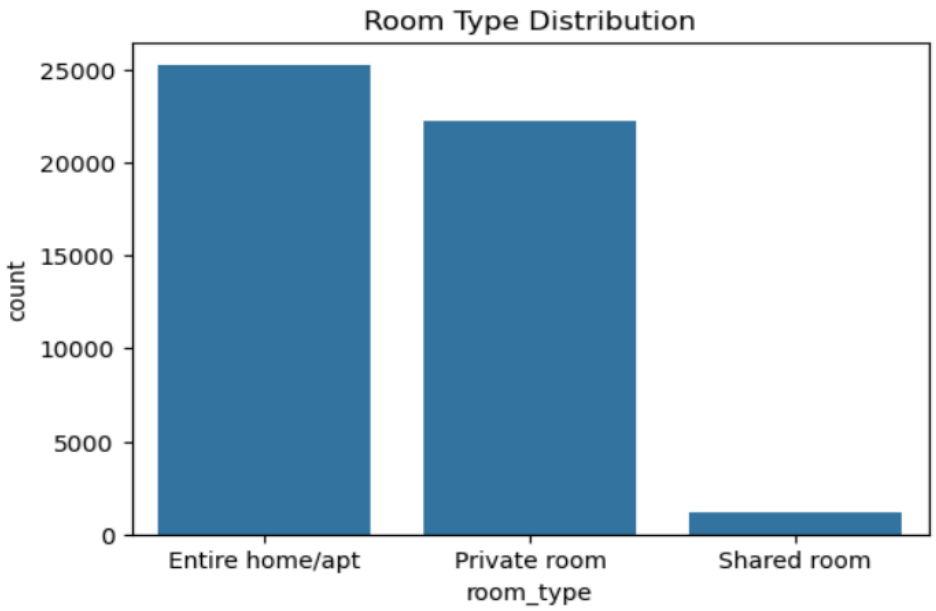

Figure 1

Figure 1: The distribution of room types revealed that entire homes and apartments constitute the majority of listings, followed by private rooms, with shared rooms representing a very small proportion of the market. This imbalance suggests that pricing dynamics are primarily driven by entire-unit listings and supports treating room type as a significant categorical predictor in subsequent modeling.

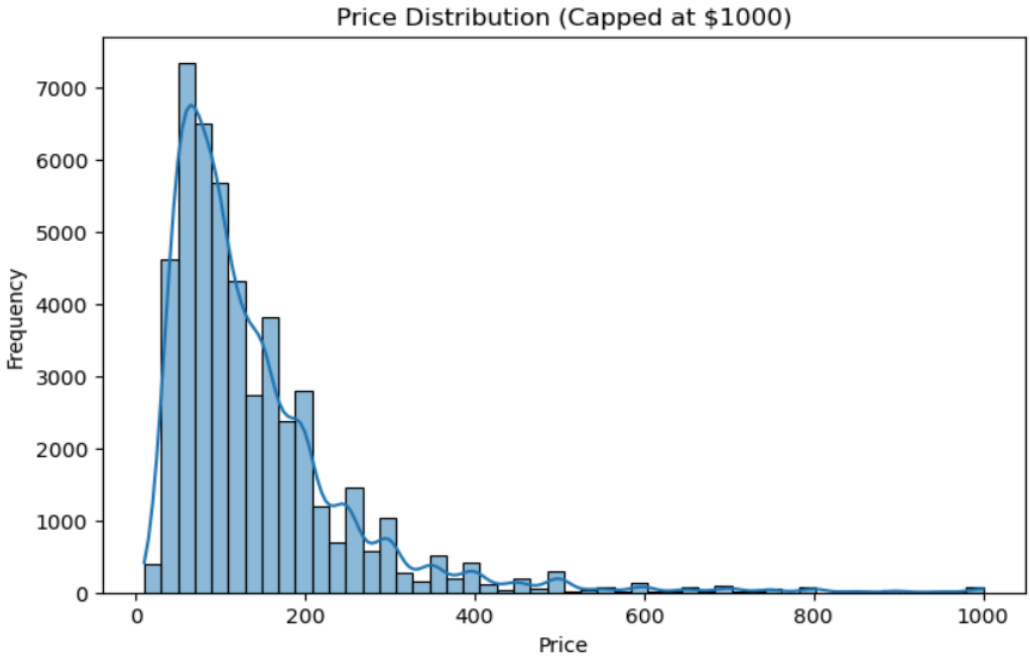

Figure 2

Figure 2: Price distribution analysis showed a strong right-skew, with the majority of listings clustered at lower price points and a long tail of high-priced properties. This pattern indicates substantial heterogeneity in listing value and justified the decision to apply a log transformation to the price variable to stabilize variance and improve model performance.

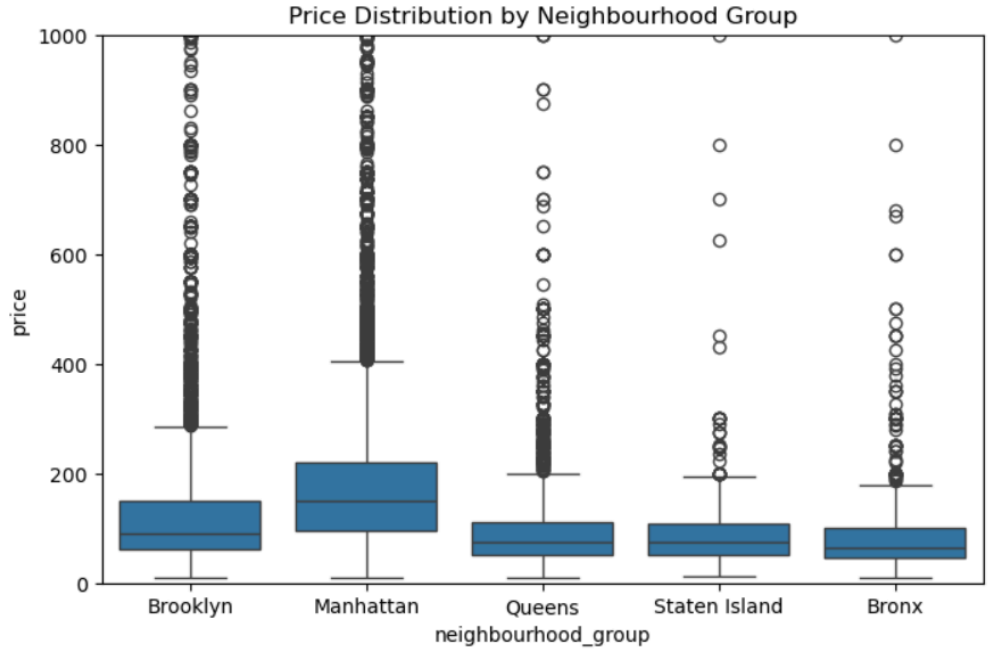

Figure 3

Figure 3: To examine spatial effects, average price by borough was visualized. The results indicate that Manhattan and Brooklyn consistently command higher average prices relative to other boroughs, reflecting differences in demand, tourism concentration, and accessibility. This finding highlights location as a dominant driver of pricing behavior.

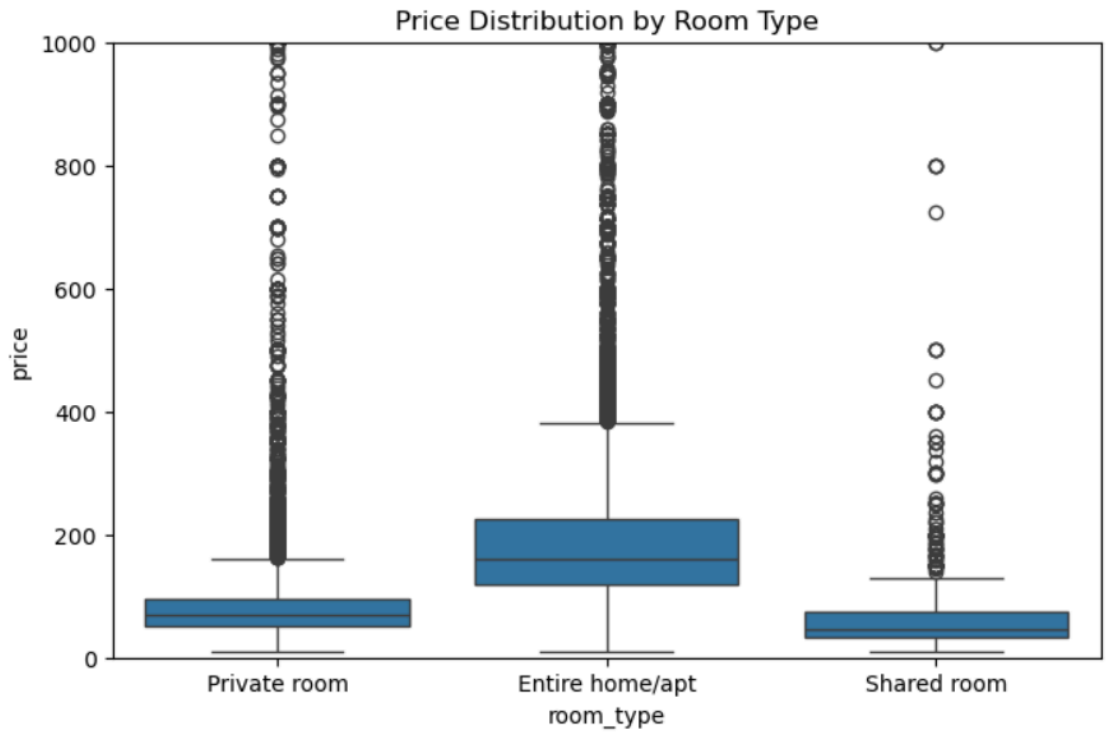

Figure 4

Figure 4: Boxplots examining price by room type further reinforced that entire homes and apartments consistently command higher prices than private or shared rooms, while also exhibiting greater variability. This variability highlights the heterogeneity within this category and supports the inclusion of room type as an important modeling feature.

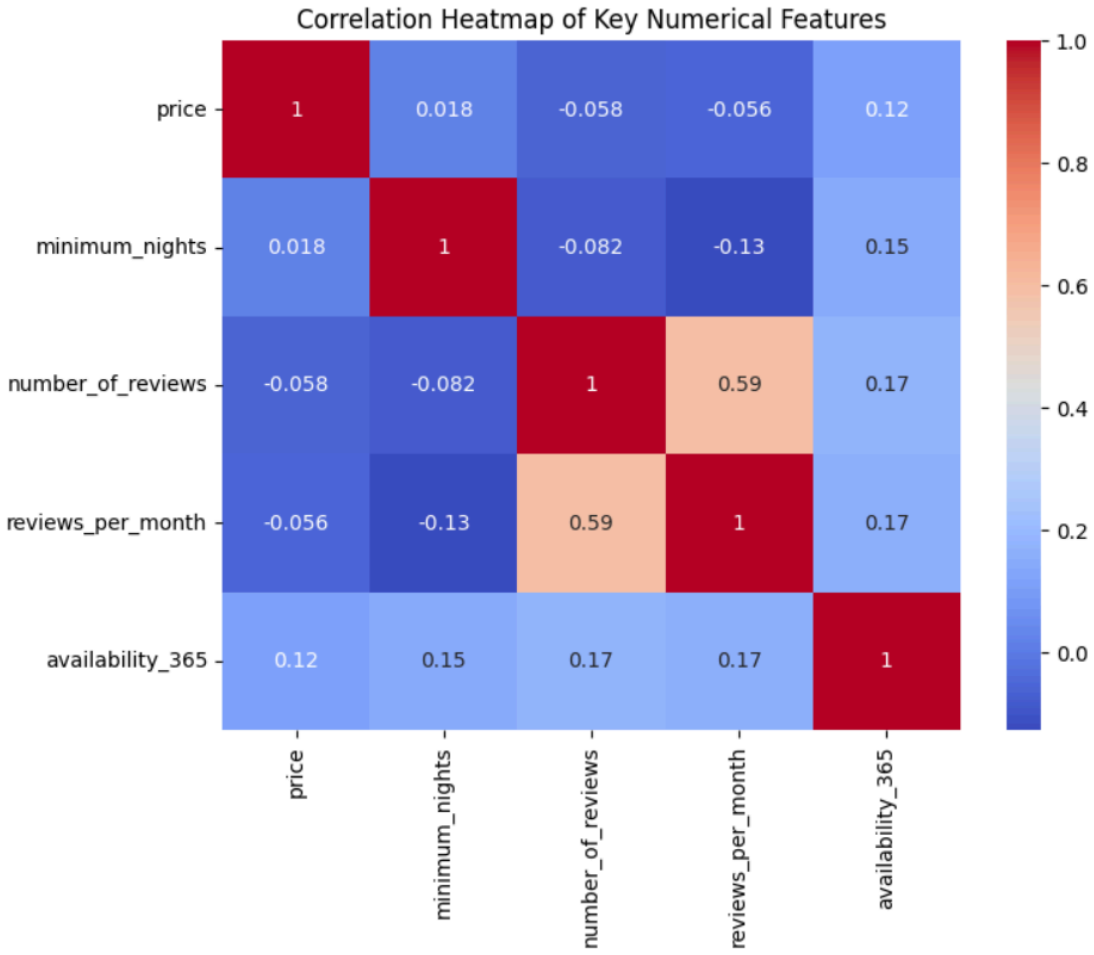

Figure 5

Figure 5: Correlation heatmaps were used to assess linear relationships among numeric variables. While no single feature showed a strong standalone correlation with price, several moderate relationships emerged—particularly between price, availability, and review-related metrics—reinforcing the need for multivariate modeling approaches rather than reliance on single predictors.

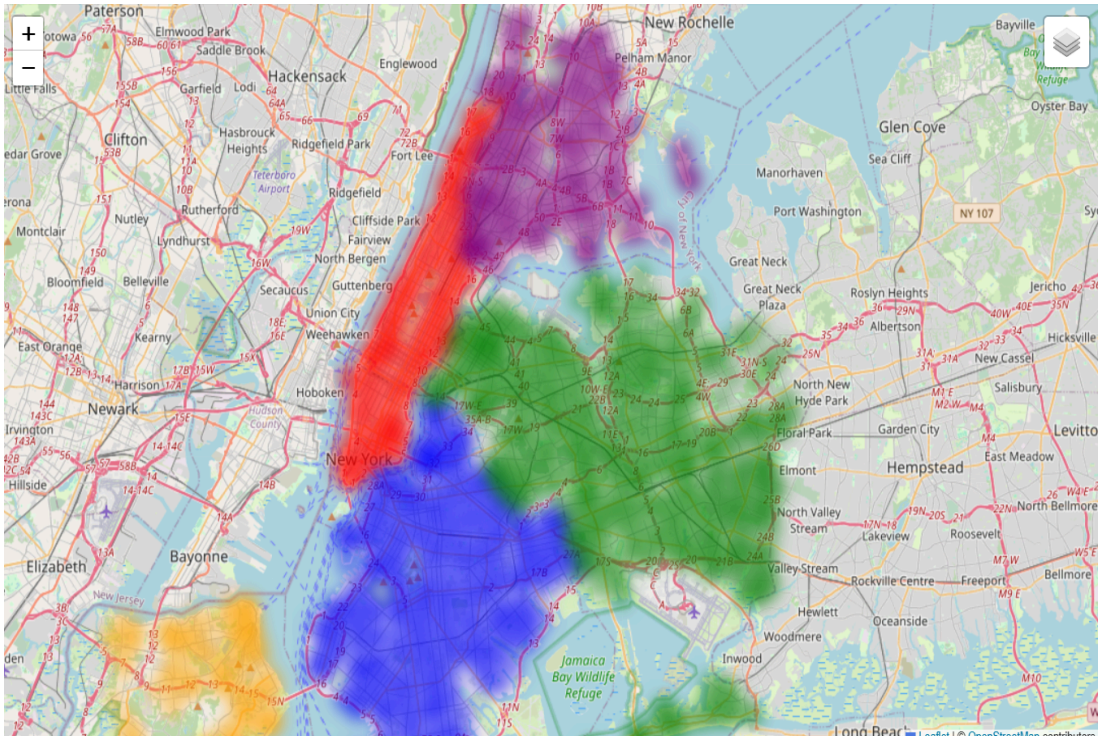

Figure 6

Figure 6: Geospatial density mapping further reinforced these patterns by illustrating clusters of high listing density and higher-priced properties concentrated in central Manhattan(red) and select areas of Brooklyn(blue). These spatial patterns suggest that proximity to key urban amenities and transit corridors plays a significant role in determining listing value.

Collectively, these exploratory analyses informed feature selection and model design by identifying the most influential variables and confirming the presence of nonlinear and spatial effects that warranted more advanced modeling techniques.

Modeling

To model Airbnb pricing behavior, several supervised learning algorithms were developed and evaluated using a structured train–test framework. The cleaned dataset was split into training and testing subsets, with model features stored in X_train and X_test, and the log-transformed price used as the target variable. This approach ensured that model performance was

evaluated on unseen data, providing a realistic assessment of generalization ability.

A baseline linear regression model was first trained on X_train to establish a reference point for performance and interpretability. While effective in capturing general trends, the linear model was limited in its ability to represent more complex, nonlinear relationships within the data.

To address this, ensemble-based models were also trained, including Random Forest and Gradient Boosting regressors. These models are better suited to capturing nonlinear interactions among features such as location, room type, and availability. All models were evaluated on X_test using standard regression metrics, including R^2, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE).

The results showed consistent improvement across models. The linear regression model achieved an R^2 of 0.585, serving as a strong baseline. The Random Forest model improved performance with an R^2 of 0.596, while the Gradient Boosting model achieved the best overall results, with an R^2 of 0.619, an MAE of approximately 0.270, and an RMSE of approximately 0.350. An R^2 value of approximately 0.62 indicates that the model is able to explain a substantial portion of the variation in Airbnb pricing using observable listing characteristics. While this does not imply perfect prediction, it demonstrates that the model captures the primary drivers of pricing behavior in the market. In practical terms, this level of performance is sufficient to support informed pricing decisions, identify under- or over-priced listings, and guide strategic adjustments. The remaining unexplained variance likely reflects factors that are difficult to quantify, such as interior design quality, host responsiveness, or hyper-local neighborhood effects. These results indicate that the Gradient Boosting model captured a greater proportion of price variability and produced more accurate predictions on unseen data.

Overall, the consistency of results across modeling approaches strengthens confidence in the findings and demonstrates that the identified pricing drivers are robust rather than model-specific.

Model	R^2	MAE	RMSE
Linear Regression	0.585	0.283	0.365
Random Forest	0.596	0.276	0.360
Gradient Boosting	0.619	0.270	0.350

Figure 7

Figure 7: Comparison of model performance across Linear Regression, Random Forest, and Gradient Boosting models. Performance is evaluated using R^2 on the test dataset (X_test). Gradient Boosting achieves the highest explanatory power, indicating superior ability to capture nonlinear relationships in Airbnb pricing data.

Findings/Takeaways

Although predicting exact Airbnb rental prices in a complex market like New York City is inherently challenging, the models developed in this project provide a reliable framework for understanding the key drivers of listing value. With an explanatory power of approximately 60%, the models are effective in identifying situations where hosts can price more aggressively, as well as cases where listings may be underpriced and leaving potential revenue unrealized. The results also clarify how specific listing characteristics—such as location, room type, and availability—directly influence expected pricing outcomes. Importantly, the modeling approach highlights actionable levers that hosts can adjust, including availability settings and listing type, to improve overall performance. While not intended to predict prices with perfect precision, these insights are sufficiently robust to inform strategic decision-making and uncover

meaningful revenue opportunities for Airbnb hosts.

Further Research

Future research could extend this analysis by incorporating temporal and seasonal trends to better capture fluctuations in Airbnb demand throughout the year. Pricing and occupancy in New York City are highly influenced by seasonality, holidays, and major events, and integrating time-based data would enable more accurate demand forecasting and dynamic pricing strategies. Additionally, performing sentiment analysis on guest reviews could provide deeper insight into how qualitative factors—such as cleanliness, communication, and location convenience—impact pricing and guest satisfaction beyond review counts alone. Finally, more advanced geospatial modeling using precise latitude and longitude data could uncover micro-location effects within neighborhoods, allowing for more granular pricing recommendations based on proximity to transit, attractions, and other amenities. Together, these enhancements would improve the predictive power of the models and provide more actionable, fine-grained insights for Airbnb hosts.